

Multiple Choice (10 marks)

1. Which compound does not contain individual molecules?
 - (a) salt
 - (b) water
 - (c) hydrogen gas
 - (d) carbon monoxide
2. Why does a floor covered with tile feel colder than a floor covered with carpet?
 - (a) The tile floor is colder than the carpet.
 - (b) The tile floor conducts heat better than the carpet.
 - (c) The carpet conducts heat better than the tile floor.
 - (d) The tile floor is a better insulator than the carpet.
3. Genes are unable to determine a person's
 - (a) eye color.
 - (b) athletic ability.
 - (c) number of teeth.
 - (d) shape of earlobes.
4. A red giant star differs from main sequence stars because it
 - (a) burns at a cooler temperature.
 - (b) is always larger than other suns.
 - (c) has greater density and gravity.
 - (d) fuses hydrogen to make helium.
5. Which is a natural cause of rivers drying up?
 - (a) pollution
 - (b) erosion
 - (c) water tables rise
 - (d) lack of precipitation

True / False (10 marks)

Answer True or False

6. True or False: The outer layers of astronauts' space suits are reflective to protect them from intense sunlight in space.
7. True or False: Landslides are a sudden change to Earth's surface caused by the movement of loose rocks downhill.
8. True or False: Humans have little control over sulfur dioxide levels, as they primarily come from natural sources like volcanoes.
9. True or False: Rubbing the coin with cold hands will make it colder than placing it in cool water.
10. True or False: Incomplete metamorphosis describes an insect life cycle where the adult stage develops without passing through a pupa phase.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the alternative energy resource that most closely resembles coal in its usage, including its benefits and challenges in comparison to traditional fossil fuels.
12. Discuss the key differences between organisms classified as Eukarya and those classified as Archaea and Bacteria, focusing on their cellular structures, functions, and ecological roles.

End of Paper